

IGOR BOBKO

Engineering Lead | Software Architect | Hands-on Full-Stack Engineer

Moscow, Russia

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Languages: Russian — Native · English — Fluent

PROFESSIONAL PROFILE

Engineering Lead and Software Architect with a strong hands-on software engineering background and more than 20 years of experience designing, building, and evolving production software systems, distributed backend platforms, enterprise integrations, web applications, and AI-enabled products.

Combine architecture ownership and engineering leadership with hands-on implementation. Experienced in defining technical direction, decomposing complex systems, coordinating delivery, distributing ownership, mentoring engineers, resolving blockers, managing technical debt, improving engineering quality, and communicating risks and trade-offs to product and business stakeholders.

Worked on international projects for Google, HarperCollins, Dow Jones, BC Technology Group, Ingram Micro, Sberbank, Beeline, and other organizations. Led and coordinated distributed engineering teams, participated in hiring and onboarding, conducted one-to-one meetings and technical reviews, and worked directly with stakeholders in the United States, Canada, and the United Kingdom.

Designed and built **Lex Humanitas**, a publicly available web and Android AI product integrating backend services, asynchronous processing, structured LLM workflows, authentication, usage accounting, billing, and resilient client applications.

Strong hands-on background in Java, Spring, Python, FastAPI, React, TypeScript, distributed systems, asynchronous workflows, cloud infrastructure, and production reliability.

ENGINEERING LEADERSHIP

Architecture and technical direction: architecture ownership, system decomposition, technical decision-making, evaluation of implementation alternatives, architecture reviews, and modernization planning.

Technical debt and system evolution: identification and assessment of technical debt, remediation prioritization, refactoring planning, legacy modernization, and balancing feature delivery with long-term maintainability.

Delivery and quality: technical planning, work decomposition, estimation, milestone tracking, dependency management, blocker resolution, code and architecture reviews, testing strategy, quality gates, release readiness, and root-cause analysis.

Reliability and risk: architectural-risk analysis, distributed-state consistency, concurrency, integration failures, retries, idempotency, failure recovery, and production reliability.

Team development: one-to-one meetings, technical mentoring, structured feedback, onboarding, technical interviews, hiring, role allocation, and ownership development.

Requirements and stakeholders: requirements discovery, ambiguity resolution, acceptance criteria, non-functional requirements, and communication of architecture, estimates, risks, and trade-offs to engineering, product, and business stakeholders.

TECHNICAL EXPERTISE

Programming languages: Java, Python, C++, JavaScript, TypeScript, SQL, Kotlin

Backend and integration: Spring Core, Spring WebFlux, Java EE, FastAPI, Node.js, REST, RPC, gRPC, SOAP, asynchronous processing, background jobs, API design, authentication, authorization

Frontend and mobile: React, Next.js, TypeScript, Vue.js, Android, Kotlin, frontend/backend integration, state management, user flows

Architecture: distributed systems, microservices, event-driven architecture, high-load and fault-tolerant systems, asynchronous workflows, state machines, concurrency, multithreading, data consistency, integration architecture

Cloud and DevOps: AWS, Microsoft Azure, Docker, Kubernetes, Terraform, CI/CD, observability, production deployment, failure recovery

Databases and messaging: PostgreSQL, MySQL, Microsoft SQL Server, Oracle, Azure SQL, Google Spanner, Redis, Kafka

AI and machine learning: OpenAI API, structured LLM outputs, schema-driven reasoning workflows, speech transcription, Hugging Face, TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV

Testing and quality: unit, integration, component, regression, contract and end-to-end testing, TDD, BDD, performance testing, root-cause analysis

PROFESSIONAL EXPERIENCE

LEX HUMANITAS — INDEPENDENT AI PRODUCT

Web: <https://humanitas.lexveritas.ai>

Android: <https://play.google.com/store/apps/details?id=ai.lexhumanitas>

Founder / Software Architect / Hands-on Full-Stack Engineer

November 2025 – Present

Public AI product for speech transcription, document analysis, communication analysis, manipulation detection, and structured LLM-based reasoning, available on web and Android.

- Designed the complete product and system architecture: backend services, asynchronous AI processing, authentication, usage accounting, billing, and resilient web and Android clients.
- Built backend services using Python, FastAPI, PostgreSQL, and Docker, and the web client using React, TypeScript, and Next.js.
- Designed the end-to-end processing pipeline from audio capture and transcription to LLM analysis, structured JSON output, persistence, and presentation.
- Designed schema-driven LLM workflows with strict JSON contracts, deterministic parsing, validation, and stable client integration, including manipulation-detection analysis of communication.
- Implemented asynchronous job processing with queues, retries, idempotency, long-running task tracking, polling, validation, and failure recovery.
- Designed hybrid authentication (device-based and account-based) and a transactional wallet with usage accounting, billing, an internal transaction ledger, and Google Play purchase validation.
- Implemented offline-resilient Android behavior with local persistence and synchronization recovery under unstable network conditions.
- Balance feature development with refactoring, reliability improvements, architectural evolution, and technical-debt reduction.
- Remain hands-on across backend, frontend, mobile integration, database design, testing, deployment, and production support.

Technologies: Python, FastAPI, PostgreSQL, Docker, React, TypeScript, Next.js, Android, Kotlin, OpenAI API, asynchronous processing, job queues, schema-driven LLM outputs

EPAM SYSTEMS

Team Lead / Senior Software Engineer

February 2021 – July 2025

Combined hands-on software development with technical leadership, architecture, requirements clarification, delivery coordination, mentoring, risk management, and stakeholder communication across international enterprise projects.

Google — GDeal Platform

San Jose, California, USA

August 2022 – July 2025

Worked within a 13-person engineering team supporting business-critical acquisition and onboarding workflows.

- Participated in technical planning, architecture discussions, implementation, and delivery coordination.
- Worked directly with Google engineers and stakeholders to clarify requirements, workflow behavior, access-control rules, integration constraints, and expected outcomes.
- Translated incomplete and ambiguous integration requirements into backend services, data-processing workflows, validation rules, access-control logic, and user-interface behavior.
- Led implementation of data-import and processing workflows, including technical decomposition, transformation, validation, failure handling, and coordination across internal systems.
- Designed and developed RPC-based backend services integrated with Google Spanner and internal Google systems.
- Developed user-interface components using Google's Wiz framework and integrated workflows with Google Sheets.
- Implemented fine-grained access control and contributed to security-sensitive integrations.
- Identified risks related to data consistency, access control, internal dependencies, integration behavior, and processing failures.
- Participated in architecture evaluation, code reviews, technical discussions, and shared engineering ownership.
- Communicated implementation alternatives, dependencies, constraints, technical risks, and delivery trade-offs.

Team: 13 people

Technologies: Java, RPC, Google Spanner, Google Sheets API, Wiz UI, CI/CD

HarperCollins Publishers — Integration and Reporting

Engineering Lead / Senior Software Engineer

February 2021 – August 2022

Led architecture and technical delivery of integration and reporting solutions during migration from legacy systems to a modernized technology environment.

- Coordinated a team of two developers and one QA engineer, distributed ownership, reviewed progress, removed blockers, and maintained delivery alignment.
- Conducted regular team meetings, one-to-one discussions, technical mentoring, architecture reviews, code reviews, and structured feedback.
- Worked directly with stakeholders in Canada and the United Kingdom.
- Clarified incomplete requirements, identified missing scenarios and contradictions, and translated stakeholder requests into technical designs, integration contracts, implementation plans, and acceptance criteria.
- Defined architecture for integration components, reusable libraries, and PDF-reporting workflows.

- Planned delivery milestones and communicated progress, dependencies, risks, assumptions, and technical trade-offs.
- Identified delivery, legacy-system, integration, and performance risks and incorporated mitigation work into the technical plan.
- Managed technical debt created by legacy constraints and balanced migration delivery with maintainability and system evolution.
- Established quality expectations for shared libraries, code reviews, regression testing, performance testing, and release readiness.
- Remained hands-on in integration development, performance analysis, reusable component implementation, and PDF-reporting solutions.

Team: 2 developers and 1 QA engineer

Technologies: Java 11, Microsoft Azure, Azure SQL, Maven, integration systems, PDF generation, performance testing

Dow Jones — Smart Search API

Team Lead / Senior Software Engineer

April 2022 – June 2022

- Coordinated technical delivery within a five-person team developing ML-powered search functionality.
- Facilitated team meetings, clarified ownership, monitored progress, supported blocker resolution, and mentored developers.
- Worked with stakeholders and engineers to clarify search behavior, API requirements, integration expectations, and technical constraints.
- Identified risks related to ML integration, concurrency, API performance, dependencies, and incomplete requirements.
- Participated in backend architecture and API design.
- Developed backend APIs for intelligent search suggestions.

Team: 5 people

Technologies: Java, REST, multithreading, dependency injection, Maven, ML-related backend integration

BC Technology Group — Crypto Trading Platform

Team Lead / Senior Software Engineer

November 2021 – February 2022

Led engineering delivery for a distributed backend platform supporting cryptocurrency margin trading.

- Coordinated a distributed five-person engineering team, assigned ownership, facilitated daily meetings, tracked delivery, and resolved blockers.
- Conducted one-to-one meetings, technical mentoring, structured feedback, interviews, hiring, onboarding, role allocation, and offboarding.
- Worked with stakeholders to clarify trading requirements, financial workflows, operational constraints, and delivery priorities.
- Participated in system architecture design and remained hands-on across backend development and DevOps.
- Managed risks related to financial correctness, concurrency, distributed state, asynchronous communication, infrastructure, and production reliability.
- Maintained engineering quality through architecture reviews, code reviews, testing expectations, and hands-on implementation.
- Balanced product delivery with technical-debt reduction, platform stability, infrastructure improvements, and architectural consistency.
- Coordinated communication between engineering, product, delivery, and infrastructure participants.

Team: 5 people

Technologies: Java, Spring WebFlux, Kafka, Redis, MySQL, AWS, Kubernetes, React

INGRAM MICRO CLOUD

Senior Software Engineer

October 2018 – February 2021

- Developed and enhanced the Odin Automation Platform for cloud-service management, customer accounts, billing, business intelligence, and user workflows.
- Worked on enterprise-grade cloud software in distributed international teams.
- Integrated platform components with Salesforce and Microsoft Azure services.
- Developed backend and frontend functionality using Java, Python, and Vue.js.
- Improved business-intelligence and reporting capabilities across platform modules.
- Developed and maintained unit, integration, component, and end-to-end tests.
- Participated in technical discussions, code reviews, production support, and platform evolution.

Technologies: Java EE, Spring, Docker, Kubernetes, PostgreSQL, Python, Vue.js, Azure SQL, Keycloak

I-TECO / SBERBANK

Senior Software Engineer — Contractor for Sberbank

August 2017 – September 2018

Developed backend services for Sberbank Online Insurance, including business workflows, auditing, logging, and integrations with internal banking systems, frontend components, and external services.

Team: 10 people

Technologies: Java EE, Spring, Jetty, IBM WebSphere, Oracle, SOAP, REST

PROGRESS SOLUTION / BEELINE

Senior Software Engineer

December 2015 – August 2016

Developed backend and frontend functionality for Beeline's ERP system used to manage mobile base-station construction.

Technologies: Java, Spring, Hibernate, JavaScript, Oracle, Informix

NOVARDIS / TEAMIDEA

Software Developer

June 2014 – June 2015

Developed SAP Hybris e-commerce solutions for Komus, Rive Gauche, and Technodom.

Technologies: SAP Hybris, Java, e-commerce integrations

VOKRUG SVETA

Programmer

February 2012 – June 2013

Developed an OpenStreetMap-based map-tile generation system.

Technologies: Python, Mapnik, OpenStreetMap

ADPO

Lead Programmer

May 2009 – October 2011

Led development of ERP and CRM systems supporting billing, accounting, customer workflows, website operations, and internal business automation.

GTKONSAL

System Administrator / Programmer

June 2005 – May 2009

Designed and developed an internal business-management system from scratch, including application architecture, database design, modular role-based functionality, runtime internationalization, secure TCP/IP access, client applications, and company websites.

Technologies: C++, TCP/IP, PHP, Microsoft SQL Server, MySQL

EDUCATION

Innopolis University — Innopolis, Russia

Master's Degree in Software Engineering | 2016–2018

Russian State Social University — Moscow, Russia

Specialist Degree in Journalism | 2007–2013

INDEPENDENT RESEARCH AND PUBLISHING

Deconstruction of Reality

Independent Researcher and Author | 2024 – Present

Website: <https://deconreality.com>

YouTube: <https://www.youtube.com/@DeconstructionOfReality>

Independent philosophical research and publishing project focused on the distinction between reality and its descriptions, facts and interpretations, form and meaning, as well as the mechanisms through which conceptual substitutions and manipulative framings shape human understanding.

Publish long-form articles and video materials developing a systematic framework for analysing interpretation, cognition, manipulation, subjectivity, and the limits of human and machine-generated models.